

PRIVATE WIRELESS NETWORKS: Reshaping Industry Connectivity And Efficiency

Explore how private wireless technology is transforming real-world industrial scenarios by overcoming operational challenges and ushering in a new era of connectivity, efficiency, and resilience through innovative solutions



The author, **ABIN CHACKO MOODHIYIL**, is Global Marketing Practice Leader, Network & Edge, Kyndryl

In the realm of modern industry, connectivity, efficiency, and reliability are paramount. The ever-accelerating pace of technological innovation has ushered in a new era, with private wireless technology standing at the forefront of this revolution. Moving beyond conventional connectivity solutions, private wireless networks offer a robust and resilient alternative, designed to meet the unique demands of various industrial sectors. From manufacturing giants in India to sprawling industrial complexes in the United States, the application of private wireless technology is redefining operational processes, emphasising its vital role in Industry 4.0.

Private wireless technology, although it might sound esoteric, isn't entirely new. Think of it as an advanced version of the cellular network we use, optimised for a confined environment. Over time, it has taken on various names, such as private

long-term evolution (LTE) or private cellular. For instance, envision an isolated network within a vast exhibition centre. In this scenario, all mobile devices communicate through this specific network rather than connecting to the broader internet. This setup exemplifies a private wireless system.

The trajectory of private wireless technology has been dynamic, transitioning from 4.9G/LTE frameworks and gravitating towards the boundless potential of 5G. The future of industry operations, particularly in the realms of safety, security, and efficiency, is deeply intertwined with the progress of private wireless networks. Additionally, an ongoing debate compares Wi-Fi and private wireless. At its core, while both facilitate wireless connectivity, private wireless offers a broader range of capabilities. It's no longer just about connectivity, but also about enhancing operational efficiencies, ensuring stringent security with built-in encryption, and adapting to modern industrial requirements.

The buzzword 'Industry 4.0' is making the rounds, signifying a paradigm shift underpinned by technological advancements like virtualisation, artificial intelligence (AI), and even quantum computing. Private wireless is at the heart of this revolution, playing a pivotal role in redefining business and operational processes. Historically, information technology (IT) dominated industrial operations. Fast forward to today, operational technology (OT), which focuses on machine-driven automation, is gaining significance. This harmonious convergence

